# ASWIN A

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## Professional Summary

Electronics and Communication Engineering student with experience in Data Analytics, Machine Learning, Backend Development, and IoT Systems. Skilled in Python, SQL, Statistical Analysis, Data Visualization, and Predictive Modeling. Built end-to-end analytics solutions involving data cleaning, feature engineering, dashboard development, and machine learning model deployment. Passionate about solving business problems through data-driven decision making.

## Education

B. Tech - Electronic and Communication Engineering — Aug 2023 – May 2027
S.R.M Institute of Science and Technology, Ramapuram, Chennai
CGPA: 8.74/10

## Technical Experience & Projects

**Global Cost of Living Analysis using Price Level Index** LINK

- Analysed 6,800+ World Bank records to study global cost-of-living and purchasing power trends using PLI data.
- Performed data cleaning, feature engineering, outlier handling, and statistical analysis using Python and Pandas.
- Built interactive Power BI dashboards with KPI cards, global maps, and country-level trend analysis.
- Developed and evaluated ML models including Logistic Regression and Random Forest with confusion matrix.

**Tech Stack:** Python, Pandas, Scikit-learn, TensorFlow, Power BI, Matplotlib

**Chennai Property Tax Analysis (2013–2018)** LINK

- Analyzed multi-year ward-level property tax collection and demand data across Chennai to identify revenue trends and collection efficiency.
- Performed data cleaning, transformation, and exploratory data analysis using Python and Pandas.
- Built interactive Power BI dashboards with zone-wise, ward-wise, and year-over-year tax collection insights.
- Developed KPI metrics and visualizations to identify high-performing and underperforming regions.
- Generated actionable insights through trend analysis and statistical exploration of residential and commercial tax data.

**Tech Stack:** Python, Pandas, NumPy, Power BI, Excel, Data Visualization

**Earthquake Auto-Shutdown System – IoT & Edge ML** LINK

- Applied a real-time seismic detection system achieving ~97% vibration classification accuracy using Edge Impulse ML models.
- Improved an end-to-end IoT pipeline transmitting sensor data and emergency alerts to the cloud via Azure IoT Hub (MQTT).
- Programmed relay-based automated safety shutdown mechanisms to immediately power down machinery during detected seismic events.
- Demonstrated practical integration of embedded systems, machine learning, and cloud communication in safety-critical scenarios.

**Tech Stack:** Embedded C/C++, ESP32, MPU6050, Edge Impulse, Machine Learning, IoT, MQTT, Azure IoT Hub

## Technical Skills

**Programming:** Python, C++, SQL, Bash

**Data Analytics:** Pandas, NumPy, Power BI, Excel

**Machine Learning:** Scikit-Learn, TensorFlow, Feature Engineering, Classification Models

**Backend Development:** Flask, REST APIs, Database Design

**Databases:** SQLite, MySQL, PostgreSQL

**Tools:** Git, GitHub, Postman, Linux, VS Code

**Soft Skills:** Problem Solving, Teamwork, Time Management, Independent Project Ownership, Self-Directed Learning

## Certifications & Activities

- Internet of Things (IoT) LINK
- Python for Data Science LINK
- Machine Learning Workshop LINK